

SAMRUDDHI KAULE

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AI/ML Developer | Full-Stack Developer

SUMMARY

Final-year Computer Science undergraduate with a strong foundation in Artificial Intelligence and Machine Learning hands-on experience in data-driven problem solving, machine learning workflows, application development, and research, supported by practical technical work. Analytical, adaptable, and self-driven, with a strong ability to learn new technologies and translate concepts into practical solutions.

SKILLS

Programming Languages: Python, SQL, C, C++

Machine Learning & AI: PyTorch, Scikit-learn, NLP, Computer Vision, Recommendation Systems

Data & Analytics: Pandas, NumPy, Feature Engineering, Power BI

Frameworks & Tools: FastAPI, Django, OpenCV, Git

Web & Databases: React, REST APIs, PostgreSQL, MongoDB

PROJECTS

Real-Time Hand Gesture Recognition System

[GitHub](#)

- Developed a real-time gesture recognition system classifying 36 alphabet/digit gestures from 42 normalized hand-landmark features, achieving 95.87% accuracy and 0.952 Macro F1 on 653 held-out samples.
- Built a position- and scale-invariant feature normalization pipeline and trained a Random Forest ensemble on 3,263 samples, achieving a 91.27 percentage-point lift over the majority-class baseline.
- Optimized inference for low-latency deployment, achieving 3.21 ms median and 5.58 ms P95 latency at 275 predictions/second on precomputed landmarks.

CineMatch — Hybrid Movie Recommendation System

[GitHub](#)

- Built a hybrid recommender using TF-IDF (Scikit-learn), SVD (Surprise), and NCF (PyTorch) to address cold-start and sparsity.
- Added context-aware personalization based on user mood and time-of-day, and used FAISS for sub-millisecond real-time similarity search.
- Developed a FastAPI backend with a React frontend and NLP-based chatbot for interaction; evaluated using RMSE (~1.04), Precision@10, and Hit Ratio@10, improving over standalone models.

RetailX — Sales Analytics

[GitHub](#)

- Built an end-to-end sales analytics pipeline processing 32K+ transactions, 20K orders, 2,999 customers, and 500 products using Python, SQL, ML, and Power BI.
- Cleaned and validated data by removing 159 duplicates, resolving 966 missing values, and correcting 65 extreme discounts, retaining 99.51% of records.
- Developed K-Means customer segmentation using 6 behavioral features, identifying 4 customer segments, including one segment representing 23.27% of customers and 38.01% of revenue.

PROFESSIONAL EXPERIENCE

AI-ML Virtual Intern — Google for Developers (EduSkills)

Apr 2025 – Jun 2025

- Performed data preprocessing, feature engineering, and model training on real-world datasets using Python.
- Applied machine learning techniques for classification and prediction tasks, evaluating performance using standard metrics.
- Implemented end-to-end ML workflows including data cleaning, model building, and result interpretation.

Data Science Virtual Internship — Siemens (EduSkills)

Apr 2026 – Jun 2026

- Completed a 10-week virtual internship covering data science, machine learning, and data analytics concepts.
- Used the RapidMiner platform for data preprocessing, visualization, model building, and evaluation through hands-on assessments.

EDUCATION

MIT-ADT University, MIT School of Computing

Expected May 2027

B.Tech in Computer Science (AI & Edge Computing) — CGPA: 8.47 / 10

CERTIFICATIONS & ADDITIONAL INFORMATION

Machine Learning I (Columbia) • AI Agents (Google, via Kaggle) • AI Principles with Edge Computing (Coursera) • Power BI Bootcamp (Nucleus, SSCBS)

Languages: English, Hindi, Marathi, Basic German

Interests: Basketball, Tennis, Reading